| Overview of whole program execution metrics | | | | |
|---|---|---|---|---|
| Number of processors | 1 | 4 | 8 | 16 |
| Elapsed time (sec) | 7.59 | 4.24 | 2.85 | 2.01 |
| Speedup | 1.00 | 1.79 | 2.66 | 3.78 |
| Efficiency | 1.00 | 0.45 | 0.33 | 0.24 |

Table 1: Analysis done on Wed Oct 28 03:38:23 PM CET 2026, cpds1101

| Overview of the Efficiency metrics in parallel fraction, $\phi$=99.63% | | | | |
|---|---|---|---|---|
| Number of processors | 1 | 4 | 8 | 16 |
| Global efficiency | 99.94% | 44.92% | 33.45% | 23.86% |
| Parallelization strategy efficiency | 99.94% | 51.31% | 43.43% | 38.80% |
| Load balancing | 100.00% | 97.83% | 95.37% | 93.68% |
| In execution efficiency | 99.94% | 52.45% | 45.54% | 41.41% |
| Scalability for computation tasks | 100.00% | 87.54% | 77.02% | 61.50% |
| IPC scalability | 100.00% | 87.97% | 78.20% | 64.73% |
| Instruction scalability | 100.00% | 99.59% | 98.63% | 95.39% |
| Frequency scalability | 100.00% | 99.92% | 99.86% | 99.60% |

Table 2: Analysis done on Wed Oct 28 03:38:23 PM CET 2026, cpds1101

| Statistics about implicit or explicit tasks in parallel fraction | | | | |
|---|---|---|---|---|
| Number of processors | 1 | 4 | 8 | 16 |
| Number of explicit tasks executed (total) | 1000.0 | 16000.0 | 64000.0 | 256000.0 |
| LB (number of explicit tasks executed) | 1.0 | 0.97 | 0.95 | 0.95 |
| LB (time executing explicit tasks) | 1.0 | 0.98 | 0.96 | 0.97 |
| Time per explicit task (average us) | 7555.55 | 538.36 | 152.36 | 47.04 |
| Overhead per explicit task (synch %) | 0.02 | 94.62 | 129.49 | 153.88 |

Table 3: Analysis done on Wed Oct 28 03:38:23 PM CET 2026, cpds1101